



ASET Competency Logbook

Candidate: Pik Fung Tai (Francis)
Discipline: Electronics Engineering Technology

ASET ID : 144xxx
As of : 2026-01-05

Work Context Details

Employer: Milo Enterprises Inc. Email: [REDACTED]
Supervisor: [REDACTED] Phone: 604 424 8228 [REDACTED]
Period: to Nov 3, 2025 to Present Title: Product Manager

Competency and Indicator Details

Information for Applicants

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IMPORTANT NOTES : Please have your supervisor review your completed Competency Summary and then sign this page to declare agreement with your level of competency. The Summary can be signed digitally or can be printed, signed, and scanned.

If you are providing multiple supervisory references to cover different positions or places of employment, each supervisor must sign a copy of this declaration. If you do not have a supervisory reference, please contact ASET at registration@aset.ab.ca to request a special consideration form.

Once you have received the signed declaration(s) from your supervisor(s), upload your Competency Summary and the signed declaration(s) to your online application portal.

Supervisor Declaration

I certify that I have direct knowledge of the applicant's work and that I have reviewed and verified that information provided in this Competency Summary to ensure it is complete, true, and accurate to the best of my knowledge.

Name : [REDACTED]
Relationship to Applicant : Supervisor
Organization Name : Milo Enterprises Inc.
Email Address : [REDACTED]
Signature : [REDACTED]

Designation : Managing Director
ASET/APEGA No. :

Date : Jan 5, 2026

Competency Summary

Employer: Milo Enterprises Inc.
Candidate: Pik Fung Tai (Francis)
ASET ID: 144 XXX
Role: Product Manager
Period: Nov 3, 2025 – Present
Date: Jan 5, 2026

(1) Indicator – Perform quantitative and qualitative analyses and tests appropriate to the discipline

Project / Work Context Title:

Perform Quantitative and Qualitative Analyses for Embedded System Functional Validation

Project Description:

I performed system-level quantitative and qualitative analyses to support functional validation of an embedded electronic product incorporating edge-AI capability. The assessment focused on key hardware subsystems, including an [REDACTED] microcontroller, camera module, and digital MEMS microphone, evaluated during early product definition and system architecture planning.

Quantitative analysis included estimating processing capacity, detection accuracy targets, expected performance margins, and preliminary BOM cost to confirm alignment with functional and commercial requirements. Qualitative analysis involved evaluating the suitability of video and audio inputs under anticipated environmental conditions and identifying technical risks related to noise sensitivity, reliability, and system accuracy. Findings were used to inform feasibility assessment and early design decisions.

(2) Indicator – Assess information to develop solutions and identify

Project / Work Context Title:

Assess Information to Develop [REDACTED] System-Level Product Specifications

Project Description:

I assessed functional requirements, hardware constraints, cost targets, and deployment considerations to develop a system-level product specification for an [REDACTED]-based embedded electronic product. This assessment defined feature scope, performance expectations, and interface requirements to support feasible early-stage product development planning.

Technical trade-offs among processing capability, detection accuracy, cost, and system performance were evaluated to ensure that proposed specifications were achievable within available resources and target BOM limits. The resulting specifications supported informed decision-making for subsequent design and development activities.

(3) Indicator – Communicate the results to the appropriate party

Project / Work Context Title:

Communicate the Result to the Appropriate Party (Meetings, Specifications and BOM Updates)

Project Description:

I communicated key technical findings and evaluation results for the [REDACTED] offline edge-AI detection system through structured meetings and updated documentation. This included presenting chipset evaluation results, detection-accuracy findings, and environmental performance issues to my supervisor. I prepared and updated technical specifications, preliminary BOMs, and kick-off documents to summarize detection limitations, processing constraints, and recommended hardware improvements. Based on these results, I communicated required system-design changes, including sensor-configuration updates and the option to add a low-cost AI co-processor to improve offline accuracy. These communications ensured management understood the technical trade-offs and requirements needed for informed decision-making and aligned next-stage development planning.

(4) Indicator – Configure hardware and software

Project / Work Context Title:

Configure Hardware and Software -

Offline Edge AI Detection System

Project Description:

I configured and optimized the hardware and software for an []-based offline edge-AI detection system during feasibility study and chipset selection. This included selecting the microprocessor with different camera and microphone options, and reviewing system limitations to evaluate chipset choices and detection-accuracy performance.

Through these configurations, I identified issues such as limited night-vision performance, rain- and wind-related interference, and reduced detection accuracy. Based on these findings, I updated hardware and software settings, selected a higher-resolution camera module, and added a low-cost AI co-processor instead of relying on an expensive cloud-based YOLO system. These configuration and hardware changes enabled stable system operation with acceptable predicted detection accuracy while maintaining efficient performance within the target BOM cost.

(5) Indicator – Explain basic electrical concepts, such as power supplies

Project / Work Context Title:

Explain Basic Electrical Concepts for AC-DC Power Supply Options

Project Description:

I applied and communicated fundamental electrical concepts related to power supplies, including AC–DC conversion principles, current and power requirements, and power-consumption considerations for IoT products.

This work involved evaluating the use of external power adapters versus built-in universal 100–240 V AC power supply options, considering electrical safety requirements, regulatory compliance (CE/UL), and thermal implications. The analysis supported system-level design decisions and management evaluation.

(6) Indicator – Describe basic electronic components, such as microprocessors, capacitors, resistors, and digital and analog circuits

Project / Work Context Title:

Describe Basic Electronic Components, Microprocessor and Circuits in an []-Based Embedded Product

Project Description:

I described and explained the function and interaction of basic electronic components used in an edge-AI IoT product, including the [] image sensor module, and a digital MEMS microphone.

I also explained the role of fundamental electronic components such as resistors and capacitors and their application within basic digital / analog circuit block diagrams to support system understanding, technical communication, and early design discussions.

(7) Indicator – Provide technical support and recommendations for upgrades and enhancements

Project / Work Context Title:

Provide Technical Support and Upgrade Recommendations for Edge-AI IoT Products

Project Description:

I provided technical support and formulated recommendations for system upgrades and enhancements to edge-AI-based IoT products utilizing [] platforms with integrated camera sensors and digital MEMS microphones.

This work involved evaluating existing hardware configurations, interface compatibility, and cost-performance constraints, as well as assessing feasible upgrade options to improve system functionality, and usability. Technical

findings, trade-off analyses, and recommendations were documented in a structured manner to support management decision-making and planning for product upgrades or new product development.

(8) Indicator – Perform calibration of electronic components and systems

Project / Work Context Title:

Perform Calibration and Verification of Electronic Components and System

Project Description:

I performed functional calibration and verification activities during the design and feasibility study stage of an edge-AI-based embedded electronic system utilizing an [REDACTED] platform. The work focused on camera modules and microphone module selection and was conducted to evaluate expected system accuracy, performance margins, and alignment with target BOM constraints.

Based on feasibility-driven mapping of input parameter ranges from the camera and microphone modules, I assessed the technical viability of the proposed system architecture and selected components to support system validation and cost considerations for subsequent product development.

(9) Indicator – Accept responsibility for professional assignments only when qualified by training and experience

Project / Work Context Title:

Accept Responsibility for Professional Assignments Only When Qualified by Training and Experience

Project Description:

I demonstrated professional responsibility by undertaking technical assignments strictly within the scope of my training, experience, and technical competence. This included verifying technical information, maintaining accurate and traceable documentation, and following established procedures to ensure accountable and compliant work practices.

When technical uncertainties or activities outside my direct expertise were identified, I escalated the matters and consulted management for guidance and decision-making. This ensured that decisions were made with appropriate technical oversight and that responsibilities were assigned to suitably qualified individuals.

(10) Indicator – Use appropriate non-verbal communication

Project / Work Context Title:

Use of Non-Verbal Communication to Support Technical Understanding

Project Description:

During benchmarking and early-stage embedded product development activities, I used appropriate non-verbal communication methods to support clear technical understanding among both engineering and non-engineering stakeholders.

This included preparing and reviewing system block diagrams, comparison tables, and BOM summaries to visually communicate system architecture, specifications, and hardware features to support effective discussions and informed commercial decision-making.



[REDACTED]
Managing Director
Milo Enterprises Inc.